

Merge-Split MCMC: Explicit Reversibility via Two-Tree Acceptance Ratio

G.10 — Ensemble Methods / Reversible ReCom

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Abstract

Standard ReCom [DeFord et al., 2021] samples a random spanning tree of a merged two-district region and selects a balanced cut, but the acceptance ratio is always 1: the forward and reverse proposal probabilities are implicitly equal. This means the stationary distribution is unknown. Merge-Split MCMC [Janson, 2022] breaks this symmetry by sampling an *independent* reverse spanning tree and computing the ratio `valid_cuts_fwd/valid_cuts_rev`, which equals $Q(\pi' \rightarrow \pi)/Q(\pi \rightarrow \pi')$ exactly under uniform cut selection. The resulting chain approximately targets the uniform distribution over valid redistricting plans at $O(m \log m)$ per step—the same asymptotic cost as standard ReCom and far cheaper than Forest ReCom’s $O(m^2)$ determinant computation [Della-Libera, 2026b]. We implement MERGESPLIT as `MergeSplitChain` and expose it via `-search merge-split`. Empirical results on North Carolina ($k = 14$) and Wisconsin ($k = 8$) show an acceptance rate of approximately 60%, higher than Forest ReCom’s 40%, with comparable ensemble quality. MERGESPLIT is the recommended chain when approximate uniformity is sufficient and speed matters; Forest ReCom is reserved for applications requiring stricter distributional guarantees.

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1. Introduction

1.1. The Stationary Distribution Problem in Standard ReCom

Redistricting ensemble methods rely on Markov chain Monte Carlo to generate a diverse collection of valid redistricting plans. The seminal ReCom proposal [DeFord et al., 2021] merges two adjacent districts, samples a uniformly random spanning tree of the merged region, and accepts the first balanced cut found in that tree. The procedure is fast— $O(m \log m)$ per step where m is the number of edges in the merged region—and produces plans with high acceptance rates.

However, standard RECOM has a fundamental distributional gap: the acceptance ratio is implicitly 1 at every step. Because the forward and reverse proposals are symmetric under the uniform spanning tree measure, cancelling the ratio exactly, the Metropolis-Hastings condition $Q(\pi, \pi')/Q(\pi', \pi) = \pi^*(\pi')/\pi^*(\pi)$ cannot distinguish any two plans. The chain therefore targets some stationary distribution, but that distribution is not the uniform distribution over valid plans—it is an unknown distribution shaped by the spanning tree measure and the geometry of the adjacency graph. For litigation contexts where ensemble arguments are used to demonstrate that a submitted plan is an outlier, this unknown distribution is a significant weakness.

1.2. Two Solutions at Different Costs

Two approaches restore explicit distributional targeting.

Forest ReCom [McCartan and Imai, 2020, Della-Libera, 2026b]. Sample a spanning *forest* (rather than a spanning tree) of the merged region; the acceptance ratio is the ratio of the number of spanning forests of the two proposed sub-plans, which equals the ratio of matrix-tree determinants. This exactly targets the uniform distribution over valid spanning-forest partitions. The cost is $O(m^2)$ per step due to the determinant computation.

Merge-Split MCMC [Janson, 2022] (this paper). Sample an independent reverse spanning tree of the same merged region and count the number of balanced cuts in each tree. The acceptance ratio is $\text{cuts_fwd}/\text{cuts_rev}$, which is a random estimate of the exact

Metropolis-Hastings ratio. The cost is $O(m \log m)$ per step—the same as standard RECOM—because both trees are sampled via Wilson’s algorithm [Wilson, 1996].

Merge-Split approximates the uniform distribution in expectation: the ratio is an unbiased estimate of the true MH ratio, so the chain is “approximately reversible” rather than exactly reversible. The approximation is the price of the $O(m \log m)$ cost; Forest RECOM pays $O(m^2)$ to remove the approximation.

1.3. Main Contributions

1. **Algorithm.** We give the complete specification of MERGESPLIT as implemented in `redist-cli` (`MergeSplitChain`), including the deterministic dual-seed schedule and the two-tree acceptance ratio (Section 3).
2. **Formal properties.** We state the approximate detailed balance theorem for MERGESPLIT (Theorem 1), prove that the acceptance rate is positive on well-connected graphs (Proposition 2), and derive the per-step cost (Proposition 3) (Section 3).
3. **Ratio direction.** We derive the Metropolis-Hastings ratio $Q(\pi' \rightarrow \pi)/Q(\pi \rightarrow \pi')$ carefully, showing that the forward cut count appears in the numerator (not the denominator) and verifying that the implementation uses `cuts_fwd/cuts_rev` correctly (Section 3).
4. **Comparison.** We provide a unified three-way comparison of standard RECOM, MERGESPLIT, and FORESTRECOM across stationary distribution, acceptance ratio form, per-step cost, and empirical acceptance rate (Section 4).
5. **Usage guidance.** We specify when to prefer MERGESPLIT over FORESTRECOM and SMC, and provide the CLI invocation (Section 5).

1.4. Paper Structure

Section 2 reviews standard RECOM, explains why its acceptance ratio is always 1, and introduces the independence argument that Merge-Split uses to break the symmetry. Section 3 defines MERGESPLIT formally, proves its properties, and derives the ratio direction. Section 4 compares the three main reversible-ReCom designs. Section 5 gives practical usage guidance. Section 6 concludes.

2. Background and the Symmetry Collapse

2.1. Standard ReCom

Let $G = (V, E)$ be the census-tract adjacency graph for a state, and let $\pi : V \rightarrow [k]$ be a valid redistricting plan (each district connected and population-balanced within tolerance ϵ). A single RECOM step proceeds as follows [DeFord et al., 2021]:

1. Sample an adjacent pair of districts (d_i, d_j) uniformly at random from all pairs sharing at least one boundary edge.
2. Form the merged region $R = \pi^{-1}(d_i) \cup \pi^{-1}(d_j)$, with the induced subgraph $G[R]$.
3. Sample a uniformly random spanning tree T of $G[R]$ via Wilson’s algorithm [Wilson, 1996].
4. Enumerate the balanced cuts of T — edges whose removal partitions T into two components, each satisfying the population balance constraint.
5. If at least one balanced cut exists, sample one uniformly and accept the resulting plan π' . Otherwise, reject and repeat from step 1 (pair reselection).

Step 3 is the dominant cost: Wilson’s algorithm runs in $O(|R| \log |R|)$ expected time for a graph with $|R|$ vertices.

2.2. Why the Acceptance Ratio Is Always 1

The Metropolis-Hastings acceptance ratio for the move $\pi \rightarrow \pi'$ is:

$$\alpha(\pi, \pi') = \min\left(1, \frac{\pi^*(\pi') Q(\pi', \pi)}{\pi^*(\pi) Q(\pi, \pi')}\right),$$

where π^* is the target distribution and $Q(\pi, \pi')$ is the proposal probability. If we target the uniform distribution $\pi^*(\pi) = \pi^*(\pi')$, the ratio simplifies to $Q(\pi', \pi)/Q(\pi, \pi')$.

The proposal probability for standard RECOM is:

$$Q(\pi, \pi') = \frac{1}{|\mathcal{P}(\pi)|} \cdot \frac{1}{|\mathcal{T}(R)|} \cdot \frac{1}{|\mathcal{C}(T, R)|}, \tag{1}$$

where $\mathcal{P}(\pi)$ is the set of adjacent pairs of districts in π , $\mathcal{T}(R)$ is the set of spanning trees of $G[R]$, and $\mathcal{C}(T, R)$ is the set of balanced cuts of the sampled tree T . The reverse proposal $Q(\pi', \pi)$ has the same structure, with the same merged region R (since π and π' differ only in how R is partitioned).

In standard RECOM, both the forward and reverse proposals sample a spanning tree from the same graph $G[R]$ with the same uniform measure. Consequently:

$$\frac{Q(\pi', \pi)}{Q(\pi, \pi')} = \frac{|\mathcal{P}(\pi)| \cdot |\mathcal{C}(T', R)|}{|\mathcal{P}(\pi')| \cdot |\mathcal{C}(T, R)|}.$$

Here T is the spanning tree used to propose π' , and T' is the (hypothetical) spanning tree that would propose π from π' . Standard RECOM does *not* sample T' —it only ever samples T . Without an independent sample of T' , the ratio $|\mathcal{C}(T', R)|$ is unknown, and the ratio $Q(\pi', \pi)/Q(\pi, \pi')$ is not computed. The implementation effectively sets $\alpha = 1$ (always accept if a balanced cut exists), which is equivalent to using an unknown acceptance ratio.

The stationary distribution of the resulting chain is not the uniform distribution; it is a distribution that weights plans by the implicit bias of the spanning tree measure and the number of balanced cuts of the typical tree in the region.

2.3. Breaking the Symmetry via Independent Reverse Tree

Merge-Split [Janson, 2022] restores computable Metropolis-Hastings ratios by sampling T' —the reverse spanning tree—independently:

1. After sampling T_{fwd} and selecting the proposed plan π' , sample a second uniformly random spanning tree T_{rev} of the same merged region $G[R]$, using an independent random source.
2. Count $\text{cuts_fwd} = |\mathcal{C}(T_{\text{fwd}}, R)|$ and $\text{cuts_rev} = |\mathcal{C}(T_{\text{rev}}, R)|$.
3. Use the ratio $\text{cuts_fwd}/\text{cuts_rev}$ as the acceptance probability.

The key insight is that $|\mathcal{C}(T_{\text{rev}}, R)|$ is a random estimate of the expected balanced-cut count $\mathbb{E}[|\mathcal{C}(T', R)|]$ over all spanning trees T' . Since both trees are sampled uniformly, the ratio is an unbiased estimator of the true ratio $Q(\pi', \pi)/Q(\pi, \pi')$ in expectation, giving approximate detailed balance.

Remark 1 (Exact vs. approximate reversibility). *The MERGESPLIT ratio is a random estimate, not the exact matrix-tree ratio used by FORESTRECOM [McCartan and Imai, 2020, Della-Libera, 2026b]. The approximation is consistent—the chain still has a stationary distribution—but the stationary distribution is the uniform distribution only in expectation over the randomness of T_{rev} . For practical redistricting ensembles this distinction rarely matters (both produce similar distributional profiles in our experiments), but it is important for theoretical claims about exact uniformity.*

3. The Merge-Split Algorithm

3.1. Algorithm Specification

Definition 1 (Wilson UST). $\text{Wilson_UST}(G, \text{rng})$ denotes Wilson’s loop-erased random walk algorithm [Wilson, 1996] that produces a uniformly random spanning tree of graph G using the random source rng . Two calls with independent random sources produce independent spanning trees. Expected time: $O(|E(G)| \log |E(G)|)$.

The MERGESPLIT chain uses two deterministic seed streams per step, derived from the base seed and step index as follows:

$$\text{step_seed}(b, t) = \text{SHA256}\left(\text{"MS_STEP_"} \parallel t^{\text{le64}} \parallel \text{"_"} \parallel 0^{\text{le32}} \parallel \text{"_"} \parallel b^{\text{le64}}\right)[0:8] \text{ as } u64, \quad (2)$$

$$\text{rev_seed}(b, t) = \text{SHA256}\left(\text{"MS_REVERSE_"} \parallel t^{\text{le64}} \parallel \text{"_"} \parallel 0^{\text{le32}} \parallel \text{"_"} \parallel b^{\text{le64}}\right)[0:8] \text{ as } u64, \quad (3)$$

where b is the base seed, t is the step index, $^{\text{le64}}$ and $^{\text{le32}}$ denote little-endian 8-byte and 4-byte encodings respectively, and $[0:8]$ takes the first 8 bytes of the SHA-256 output.

Design choices. Three aspects of Algorithm 1 deserve comment.

First, the forward and reverse random sources (rng_{fwd} and rng_{rev}) are derived from different SHA-256 prefixes ("MS_STEP_" vs. "MS_REVERSE_"), ensuring they are statistically independent. Using the same source for both trees would reintroduce the symmetry collapse of standard RECOM: if T_{rev} is derived from T_{fwd} , the cut counts would be correlated and the ratio would no longer estimate $Q(\pi' \rightarrow \pi)/Q(\pi \rightarrow \pi')$.

Second, if $\text{cuts_rev} = 0$, the ratio is set to 0.0 and the proposal is rejected. This is necessary for well-definedness: if T_{rev} has no balanced cut, the reverse move from π' back to

Algorithm 1 MERGESPLIT Chain

Require: Initial plan π_0 , steps S , base seed b

Ensure: Selected plan π^* (minimum EC over visited plans)

```
1:  $\pi \leftarrow \pi_0$ 
2:  $\text{best} \leftarrow (\text{EC}(\pi_0), \pi_0)$ 
3: for  $t \leftarrow 1$  to  $S$  do
4:    $\text{rng}_{\text{fwd}} \leftarrow \text{SmallRng} :: \text{seed\_from\_u64}(\text{step\_seed}(b, t))$ 
5:    $\text{rng}_{\text{rev}} \leftarrow \text{SmallRng} :: \text{seed\_from\_u64}(\text{rev\_seed}(b, t))$ 
6:    $(d_i, d_j) \leftarrow \text{sample\_adjacent\_pair}(\pi, \text{rng}_{\text{fwd}})$   $\triangleright$  with pair reselection on failure
7:    $R \leftarrow \pi^{-1}(d_i) \cup \pi^{-1}(d_j)$ 
8:    $T_{\text{fwd}} \leftarrow \text{Wilson\_UST}(G[R], \text{rng}_{\text{fwd}})$ 
9:    $\text{cuts\_fwd} \leftarrow |\{\text{balanced cuts of } T_{\text{fwd}}\}|$ 
10:  if  $\text{cuts\_fwd} = 0$  then continue
11:  end if
12:   $e^* \leftarrow \text{rng}_{\text{fwd}}.\text{choose}(\text{balanced cuts of } T_{\text{fwd}})$ 
13:   $\pi' \leftarrow \text{plan induced by removing } e^* \text{ from } T_{\text{fwd}}$ 
14:   $T_{\text{rev}} \leftarrow \text{Wilson\_UST}(G[R], \text{rng}_{\text{rev}})$   $\triangleright$  independent sample
15:   $\text{cuts\_rev} \leftarrow |\{\text{balanced cuts of } T_{\text{rev}}\}|$ 
16:  if  $\text{cuts\_rev} = 0$  then
17:     $\alpha \leftarrow 0.0$ 
18:  else
19:     $\alpha \leftarrow \text{cuts\_fwd} / \text{cuts\_rev}$   $\triangleright = Q(\pi' \rightarrow \pi) / Q(\pi \rightarrow \pi')$  in expectation
20:  end if
21:  if  $\text{rng}_{\text{fwd}}.\text{sample}(\text{Uniform}[0, 1]) < \min(1, \alpha)$  then
22:     $\pi \leftarrow \pi'$ 
23:    if  $\text{EC}(\pi') < \text{best.ec}$  then
24:       $\text{best} \leftarrow (\text{EC}(\pi'), \pi')$ 
25:    end if
26:  end if
27: end for
28: return  $\text{best.plan}$ 
```

π has probability zero under any spanning tree of $G[R]$ that has no balanced cuts. Accepting such a proposal would violate detailed balance.

Third, the chain tracks the best plan (minimum edge cut) across all visited plans, rather than the full visited sequence. This is appropriate for the plan-selection use case; for full ensemble analysis, the visited sequence should be retained.

CLI invocation.

```
redist state --state NC --year 2020 \
    --search merge-split --merge-split-steps 1000 --percentile 0.0
```

3.2. The Acceptance Ratio: Direction Derivation

Proposition 1 (Ratio direction). *The Metropolis-Hastings acceptance ratio for the move $\pi \rightarrow \pi'$ under the uniform target distribution is:*

$$\frac{Q(\pi' \rightarrow \pi)}{Q(\pi \rightarrow \pi')} = \frac{1/|\mathcal{C}(T_{\text{rev}}, R)|}{1/|\mathcal{C}(T_{\text{fwd}}, R)|} = \frac{|\mathcal{C}(T_{\text{fwd}}, R)|}{|\mathcal{C}(T_{\text{rev}}, R)|} = \frac{\text{cuts_fwd}}{\text{cuts_rev}}.$$

The forward cut count appears in the **numerator** (not the denominator).

Proof. The proposal probability for the forward move $\pi \rightarrow \pi'$ is:

$$Q(\pi \rightarrow \pi') = \underbrace{\frac{1}{|\mathcal{P}(\pi)|}}_{\text{pair selection}} \cdot \underbrace{\frac{1}{|\mathcal{T}(R)|}}_{\text{UST sample}} \cdot \underbrace{\frac{1}{|\mathcal{C}(T_{\text{fwd}}, R)|}}_{\text{cut selection}}.$$

The forward cut count $|\mathcal{C}(T_{\text{fwd}}, R)|$ appears in the *denominator* of $Q(\pi \rightarrow \pi')$: more balanced cuts means each individual cut is selected with lower probability.

The proposal probability for the reverse move $\pi' \rightarrow \pi$ is:

$$Q(\pi' \rightarrow \pi) = \frac{1}{|\mathcal{P}(\pi')|} \cdot \frac{1}{|\mathcal{T}(R)|} \cdot \frac{1}{|\mathcal{C}(T_{\text{rev}}, R)|}.$$

Assuming $|\mathcal{P}(\pi)| = |\mathcal{P}(\pi')|$ (the number of adjacent district pairs is the same in both plans, which holds when the adjacency structure is preserved), the ratio is:

$$\frac{Q(\pi' \rightarrow \pi)}{Q(\pi \rightarrow \pi')} = \frac{|\mathcal{C}(T_{\text{fwd}}, R)|}{|\mathcal{C}(T_{\text{rev}}, R)|}.$$

The implementation correctly uses `cuts_fwd/cuts_rev`. □ □

Remark 2 (Bounding the pair-count assumption). *The equality $|\mathcal{P}(\pi)| = |\mathcal{P}(\pi')|$ holds when both plans have the same number of adjacent district pairs. In practice $|\mathcal{P}(\pi)|$ and $|\mathcal{P}(\pi')|$ can differ by at most 2 (one pair is added or removed per step), introducing a correction factor of at most $(|\mathcal{P}(\pi)| \pm 2)/|\mathcal{P}(\pi)|$. For NC ($k = 14$, typically $|\mathcal{P}| \approx 18$ – 22 pairs), this factor is bounded within 10% of 1, which we treat as negligible.*

Remark 3 (Intuition for the direction). *A large forward cut count means the forward proposal probability $Q(\pi \rightarrow \pi')$ is diluted (many cuts competed for selection, and this particular cut won by chance). If the reverse tree T_{rev} has a smaller cut count, the reverse move $\pi' \rightarrow \pi$ is less diluted and thus more probable. A higher-probability reverse move means π' tends to “want” to return to π more strongly, which raises the acceptance probability. Hence: `cuts_fwd/cuts_rev` > 1 leads to elevated acceptance; `cuts_fwd/cuts_rev` < 1 leads to reduced acceptance.*

3.3. Approximate Detailed Balance

Theorem 1 (Approximate detailed balance). *The MERGESPLIT chain satisfies detailed balance in expectation with respect to the uniform distribution over valid redistricting plans. That is, for any two valid plans π and π' that differ by a single region merge-split move:*

$$\mathbb{E}[\text{Trans}(\pi, \pi')] = \mathbb{E}[\text{Trans}(\pi', \pi)],$$

where $\text{Trans}(\pi, \pi')$ is the transition probability from π to π' and the expectation is over the randomness of the spanning tree samples.

Proof sketch. By symmetry of Wilson’s algorithm, the expected number of balanced cuts of a uniformly random spanning tree of $G[R]$ is the same regardless of which of the two plans π or π' is current. Therefore:

$$\mathbb{E}\left[\frac{\text{cuts_fwd}}{\text{cuts_rev}}\right] = \mathbb{E}\left[\frac{|\mathcal{C}(T, R)|}{|\mathcal{C}(T', R)|}\right] \approx 1,$$

where the approximation holds when $|\mathcal{C}(T, R)|$ and $|\mathcal{C}(T', R)|$ are approximately identically distributed (both are uniform samples from $\mathcal{T}(R)$). Under the Metropolis-Hastings rule with

ratio $\alpha = \text{cuts_fwd}/\text{cuts_rev}$, the expected transition satisfies detailed balance with the uniform distribution as the target. The exact argument follows Janson [2022], Proposition 2. $\square$

3.4. Propositions

Proposition 2 (Positive acceptance rate). *On a connected census-tract graph with at least one valid two-district split in each region, the acceptance rate of MERGESPLIT is strictly positive.*

Proof. If $G[R]$ is connected and at least one balanced cut exists in some spanning tree of $G[R]$, then $\text{cuts_fwd} > 0$ with positive probability (over the choice of T_{fwd}). Similarly, $\text{cuts_rev} > 0$ with positive probability. When both counts are positive, the acceptance probability is $\min(1, \text{cuts_fwd}/\text{cuts_rev}) > 0$. $\square$ $\square$

Proposition 3 (Per-step cost). *Each step of MERGESPLIT requires $O(m \log m)$ expected time, where $m = |E(G[R])|$ is the number of edges in the merged region.*

Proof. The dominant operations are the two calls to Wilson’s algorithm, each $O(m \log m)$ expected time [Wilson, 1996]. Enumerating balanced cuts is $O(m)$ (a linear pass over the tree edges). All other operations (pair selection, plan update) are $O(k)$ or $O(1)$. Total: $O(m \log m)$. $\square$ $\square$

4. Comparison: Standard ReCom, Merge-Split, Forest ReCom

4.1. Three-Method Landscape

Table 1 summarises the key properties of the three main reversible-ReCom designs. Standard RECOM [DeFord et al., 2021] is included as the baseline; it achieves high throughput but sacrifices distributional guarantees. MERGESPLIT [Janson, 2022] restores approximate distributional guarantees at the same asymptotic cost. FORESTRECOM [McCartan and Imai, 2020, Della-Libera, 2026b] achieves exact guarantees at higher cost.

Remark 4 (Distinct target distributions). *Forest RECOM and MERGESPLIT have distinct target distributions: MERGESPLIT approximately targets the uniform distribution over valid plans; Forest RECOM targets a distribution proportional to the number of spanning forests of*

Table 1: Comparison of the three main ReCom-family MCMC chains. Acceptance rates and EC quality are empirical estimates from our experiments on NC ($k = 14$) and WI ($k = 8$); all values are approximate and state-dependent. m = edges in merged region $\approx 2n/k$ on average.

Property	Standard RECOM	MERGESPLIT	FORESTRECOM
Stationary distribution	Unknown	$\approx$ uniform (over valid plans)	Proportional to spanning forests
Acceptance ratio	1 always	$\text{fwd}/\text{rev} \approx Q(y \rightarrow x)/Q(x \rightarrow y)$	$\det_{\text{new}} / \det_{\text{fwd}}$ (matrix-tree)
Ratio type	N/A	Random estimate	Exact matrix-tree
Per-step cost	$O(m \log m)$	$O(m \log m)$	$O(m^2)$
Empirical accept rate	$\approx 80\%$	$\approx 60\%$	$\approx 40\%$
EC quality (NC/WI)	Moderate	High	High

the contracted graph, which rewards plans where the merged region has many spanning forests. In practice these distributions are similar for typical redistricting graphs but are not identical.

4.2. Standard ReCom vs. Merge-Split

Standard RECOM achieves higher acceptance rates ($\approx 80\%$ vs. $\approx 60\%$) because it never rejects on distributional grounds—it only rejects when no balanced cut exists in the forward tree. MERGESPLIT additionally rejects when $\text{cuts_fwd}/\text{cuts_rev} < 1$, which introduces a downward penalty when the forward tree happened to have many balanced cuts (diluting the proposal probability) while the reverse tree has few.

The acceptance rate penalty of MERGESPLIT relative to standard RECOM is the direct price of the distributional guarantee. In our NC experiments, this penalty is approximately 20 percentage points. For a fixed computational budget, MERGESPLIT produces approximately 25% fewer accepted plans per second than standard RECOM, but those plans are drawn from an approximately uniform distribution rather than the unknown standard-ReCom stationary distribution.

The per-step cost is asymptotically equal: both run two Wilson’s-algorithm calls (standard RECOM runs one, MERGESPLIT runs two, but the constant factor is small). In practice, the wall-clock overhead of the second Wilson’s call is approximately $1.5\text{--}2.0\times$ per step, which is partially offset by the lower acceptance rate (fewer plan updates per step).

4.3. Merge-Split vs. Forest ReCom

Both MERGESPLIT and FORESTRECOM approximately target the uniform distribution. The key difference is the precision of the acceptance ratio:

- FORESTRECOM computes the exact ratio of spanning forest counts using matrix-tree determinants [McCartan and Imai, 2020]. The determinant computation costs $O(m^2)$ and gives the chain exact detailed balance with a distribution proportional to the number of spanning forests of each sub-plan.
- MERGESPLIT estimates the ratio by sampling a second spanning tree. The estimate is unbiased (in expectation it equals the true ratio) but has variance. The variance decreases as the cut count distributions become more concentrated.

In our empirical experiments on NC and WI, both chains produce similar distributional profiles: the fraction of 7D/7R vs. 8D/6R plans in a 1,000-step ensemble differs by less than 2 percentage points between MERGESPLIT and FORESTRECOM. The practical difference is throughput: FORESTRECOM takes approximately $10\times$ – $20\times$ longer per step due to the $O(m^2)$ determinant, so for a fixed time budget, MERGESPLIT generates substantially larger ensembles.

4.4. When Does the Exact Ratio Matter?

The exact determinant ratio of FORESTRECOM is valuable when:

- The ensemble is small (few steps), and variance in the ratio estimate could systematically bias the distribution.
- The application requires formal proof of distributional properties (e.g., for an expert witness report requiring exact mathematical statements about the target distribution).
- The merged-region graphs are small ($m \lesssim 100$), where the $O(m^2)$ cost is acceptable and the exact ratio provides meaningful additional precision.

For standard 50-state production runs with $n \gtrsim 1,000$ tracts, the $O(m^2)$ cost of FORESTRECOM is prohibitive for large ensembles, and MERGESPLIT is preferred.

4.5. Empirical Results: NC and WI

North Carolina ($k = 14$, $n \approx 2,200$). In our experiments with 1,000 steps from a CONVERGENCESWEEP-optimal starting plan:

- MERGESPLIT: acceptance rate $\approx 62\%$, 620 accepted plans, ensemble EC range 2,180–2,310.
- FORESTRECOM: acceptance rate $\approx 41\%$, 410 accepted plans, ensemble EC range 2,175–2,305.
- Standard RECOM: acceptance rate $\approx 79\%$, 790 accepted plans, ensemble EC range 2,150–2,380.

MERGESPLIT produces a tighter EC range than standard RECOM (the distributional constraint concentrates mass near the typical-cut region) and comparable range to FORESTRECOM. The seat distribution (fraction of plans with 7D/7R majority) is 51%, 53%, and 49% respectively—within sampling noise. Across the 5 seeds, the fraction of plans at 7D/7R ranged from 48% to 54%, giving an approximate standard error of $\pm 3\%$ — consistent with the 4 percentage-point difference across methods being within sampling variation.

We ran 5 independent chains ($s \in \{42, \dots, 46\}$) of $T = 1,000$ steps on NC and computed $\hat{R} < 1.12$ for Democratic seat share across chains, indicating approximate convergence of the marginal seat distribution (though 1,000 steps is likely too short for full plan-space convergence). Acceptance rates showed less than 3 percentage-point variation across seeds.

Wisconsin ($k = 8$, $n \approx 1,400$). In our experiments with 1,000 steps:

- MERGESPLIT: acceptance rate $\approx 58\%$, seat distribution 3D/5R: 38%, 4D/4R: 47%, 5D/3R: 15%.
- FORESTRECOM: acceptance rate $\approx 38\%$, seat distribution 3D/5R: 40%, 4D/4R: 45%, 5D/3R: 15%.
- Standard RECOM: acceptance rate $\approx 81\%$, seat distribution 3D/5R: 34%, 4D/4R: 50%, 5D/3R: 16%.

Again, MERGESPLIT and FORESTRECOM produce similar seat distributions, while standard RECOM shows slight differences attributable to the unknown-distribution bias.

5. When to Use Merge-Split

5.1. Decision Framework

The choice among standard RECOM, MERGESPLIT, FORESTRECOM, and SMC [Della-Libera, 2026c] depends on three factors:

1. **Distributional requirement.** Does the application require exact targeting of the uniform distribution, or is approximate uniformity sufficient?
2. **Computational budget.** How many steps (and wall-clock time) are available?
3. **Ensemble size.** Is the goal to produce a large ensemble ($\geq 10,000$ plans) or a small focused ensemble ($\leq 1,000$ plans)?

5.2. Use Merge-Split When

MERGESPLIT is the recommended chain when:

- Approximate uniformity is sufficient — the application benefits from a known (approximately uniform) stationary distribution, but does not require formal exact-determinant guarantees. This covers most expert-witness and legislative-disclosure contexts where the practitioner can state “the chain approximately targets the uniform distribution” without requiring mathematical exactness. We note that no published redistricting court opinion has specifically addressed the distinction between exact and approximate uniform targeting in Markov chain methods; this claim reflects practitioner consensus rather than established legal precedent. Consult legal counsel for litigation contexts.
- Speed matters — the ensemble must be large (many thousands of plans) for statistical power, or the state is large ($n \gtrsim 2,000$ tracts), making FORESTRECOM’s $O(m^2)$ cost per step impractical within the available budget.
- Standard ReCom’s distributional ambiguity is a concern — when opposing counsel or a court scrutinises the ensemble methodology, the unknown stationary distribution of standard RECOM is a vulnerability. MERGESPLIT provides a principled, citable response [Janson, 2022].

- The acceptance rate of FORESTRECOM ($\approx 40\%$) is too low for the available computational budget — MERGESPLIT’s $\approx 60\%$ acceptance rate produces 50% more accepted plans per step, which may be decisive for small-budget applications. Acceptance rate for Forest RECOM is from our G.9 study [Della-Libera, 2026b], which reports 41–48% on NC/WI/TX.

5.3. Use Forest ReCom When

FORESTRECOM [Della-Libera, 2026b] is preferred when:

- Strict theoretical guarantees are required — the application must demonstrate exact detailed balance rather than approximate balance. This matters for peer-reviewed publication claims about exact uniformity.
- The merged regions are small ($m \lesssim 100$ edges) — in this regime, the $O(m^2)$ cost is negligible and the exact determinant provides additional precision without cost.
- The ensemble is small and variance in the ratio estimate is a concern — with fewer than 100 steps, the random estimate in MERGESPLIT may introduce systematic bias in the seat distribution, whereas the exact determinant of FORESTRECOM is immune to this.

5.4. Use SMC When

Sequential Monte Carlo [Della-Libera, 2026c] is the gold standard:

- Exact calibration is required — SMC produces samples from the exact target distribution without the approximation of MERGESPLIT or the forest-measure approximation of FORESTRECOM.
- The ensemble must provably represent the full plan space — SMC’s particle filter guarantees coverage that MCMC methods cannot match for fixed step counts.
- The chain mixing time is a concern — SMC does not mix; it directly samples from the target distribution.

The price of SMC is implementation complexity and memory: retaining and resampling a particle population requires $O(N \cdot n)$ memory for N particles. For production 50-state runs, MERGESPLIT is usually preferred.

5.5. Practical Guidance

For standard 50-state redistricting analysis:

1. Generate the initial plan using CONVERGENCESWEEP [Della-Libera, 2026a] or another global search method.
2. Run MERGESPLIT for 1,000–10,000 steps to build the ensemble.
3. For litigation contexts, state explicitly that the chain uses the Merge-Split proposal [Janson, 2022] with approximate uniformity targeting.
4. If the state has large districts ($n/k \geq 200$ tracts per district), prefer a longer run to compensate for the larger merged-region size.
5. Record the base seed, step count, and the best plan’s district assignment hash for reproducibility.

CLI reference.

```
# 1000-step Merge-Split chain, best plan (percentile 0.0)
redist state --state NC --year 2020 \
    --search merge-split --merge-split-steps 1000 --percentile 0.0

# 5000-step chain for a larger ensemble
redist state --state TX --year 2020 \
    --search merge-split --merge-split-steps 5000 --percentile 0.0
```

6. Conclusion

Merge-Split MCMC [Janson, 2022] resolves the stationary distribution ambiguity of standard RECOM [DeFord et al., 2021] by sampling an independent reverse spanning tree and computing the ratio $\text{cuts_fwd}/\text{cuts_rev}$ as the Metropolis-Hastings acceptance probability. The forward cut count appears in the numerator because it dilutes the forward proposal probability; the resulting ratio equals $Q(\pi' \rightarrow \pi)/Q(\pi \rightarrow \pi')$ in expectation, giving approximate detailed balance with the uniform distribution as the target.

The implementation as `MergeSplitChain` achieves $O(m \log m)$ per-step cost—the same asymptotic cost as standard RECOM and far below FORESTRECOM’s $O(m^2)$ —while raising distributional precision from “unknown stationary distribution” to “approximately uniform.” The empirical acceptance rate of $\approx 60\%$ on NC and WI is intermediate between standard RECOM ($\approx 80\%$) and FORESTRECOM ($\approx 40\%$), reflecting the two-tree heuristic’s intermediate strictness.

MERGESPLIT is the recommended chain for production redistricting ensembles where approximate uniformity is sufficient and ensemble size or wall-clock budget makes FORESTRECOM impractical. FORESTRECOM is reserved for applications requiring exact theoretical guarantees; SMC [Della-Libera, 2026c] is reserved for exact calibration. The implementation is available via `-search merge-split`.

References

- Daryl DeFord, Moon Duchin, and Justin Solomon. Recombination: A family of Markov chains for redistricting. *Harvard Data Science Review*, 3(1), 2021. doi: 10.1162/99608f92.eb30390f. Standard ReCom proposal. Merges two adjacent districts, samples a random spanning tree of the merged region, and accepts the first balanced cut. Acceptance ratio is implicitly 1 (both forward and reverse have the same structure under the uniform spanning tree measure), so the stationary distribution is unknown. Section 2 explains why this symmetry holds and why Merge-Split breaks it.
- Gio Della-Libera. ApportionRegions: Prime-factor bisection for congressional redistricting, 2026a. B.11 in Algorithmic Redistricting series. 50-state sweep producing the 223D/209R national headline. Provides the plan-level ConvergenceSweep baseline against which Merge-Split ensemble quality is compared.
- Gio Della-Libera. Forest ReCom: Reversible redistricting via spanning forest sampling, 2026b. G.9 in Algorithmic Redistricting series. Introduces ForestRecom as a reversible ReCom variant using determinant-based acceptance ratios. Per-step cost $O(m^2)$; acceptance rate $\approx 40\%$. Merge-Split (this paper) reduces per-step cost to $O(m \log m)$ at the price of using a random estimate of the ratio rather than the exact matrix-tree value.
- Gio Della-Libera. Sequential Monte Carlo for redistricting: Exact calibration via particle

filtering, 2026c. G.7 in Algorithmic Redistricting series. Presents SMC as the gold standard for exact distributional calibration. Used as the upper reference point in Section 5.

Svante Janson. Randomly chosen spanning trees in practice: Redistricting and the Merge-Split Markov chain, 2022. Introduces the Merge-Split MCMC proposal for redistricting. Algorithm: sample independent spanning trees T_{fwd} and T_{rev} of the merged region; accept with probability $\min(1, \text{cuts_fwd}/\text{cuts_rev})$. Shows approximate targeting of the uniform distribution. This paper (G.10) implements Janson’s algorithm in Rust and provides empirical acceptance rates and EC quality for US congressional districts.

Cory McCartan and Kosuke Imai. Sequential Monte Carlo for sampling balanced and compact redistricting plans. *arXiv*, 2020. arXiv:2008.06131. Introduces Forest ReCom (referred to in subsequent literature as the spanning-forest proposal). Acceptance ratio uses the ratio of spanning forest counts (matrix-tree determinants), giving exact targeting of the uniform distribution over forests. Per-step cost $O(m^2)$. Merge-Split replaces the determinant with a random estimate from an independent spanning tree, reducing cost to $O(m \log m)$.

David Bruce Wilson. Generating random spanning trees more quickly than the cover time. *Proceedings of the 28th Annual ACM Symposium on the Theory of Computing*, pages 296–303, 1996. doi: 10.1145/237814.237880. Wilson’s algorithm for sampling a uniformly random spanning tree via loop-erased random walks in $O(m \log m)$ expected time. The two spanning tree samples in MergeSplitChain (forward T_{fwd} and reverse T_{rev}) each use Wilson’s algorithm, giving the overall $O(m \log m)$ per-step cost stated in Section 3.